

ANSI/ASME B73.1 process pump part item numbers, frame and group families, and size coverage: a reference dataset

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Source	Compiled from the manufacturer's own published parts pages, listed on the last page

What this dataset contains

Four tables covering ANSI/ASME B73.1 horizontal end-suction process pumps as built by YSM Pumps in two series: the G196 series, built to the published Goulds 3196 dimensional pattern and organised by frame (STX, MTX, LTX, XLTX), and the D Mark III series, built to the published Durco Mark III pattern and organised by group (1, 2, 3). The tables give the item number used for each part on a sectional drawing, the frame or group families and the sizes each covers, which item numbers appear on which sectional sheet, and the size designations with this manufacturer's casing and impeller references.

Why it is published

Item numbering on B73.1 pumps is widely used and rarely written down in a machine-readable form. Maintenance planners, storeroom staff and anyone reading a sectional drawing need to move between an item number, the part it identifies, and the frame or group it belongs to. This dataset states that mapping for one manufacturer's products, so it can be cited, checked and corrected.

Status and limitations

- This dataset describes YSM Pumps' own products. Goulds, 3196, Durco and Mark III are trademarks of their respective owners; they are named only to identify the published dimensional pattern a series is built to. YSM Pumps is an independent manufacturer, not affiliated with, sponsored by or endorsed by ITT Goulds Pumps or Flowserve.
- No OEM part numbers are included in this dataset. Item numbers are the standard item and POS numbering used on ANSI/ASME B73.1 sectional drawings; the YSM reference suffix is this manufacturer's own.
- Interchange applies to a pump of the same size designation and frame or group, confirmed against the nameplate and the installed pump drawing. Nothing here states that a part will fit a specific installed pump.
- For screening and identification. Values are catalogue ranges, not guarantees for every size and configuration; a quotation is confirmed against size-specific reviewed documents.

Method and Field Definitions

How the tables were compiled, and what each column means.

Method

Every value was extracted programmatically from tables already published on the manufacturer's own website on 2026-09-22, not transcribed from drawings or scanned documents. The extraction script reads the HTML tables, normalises the size designation separator to 'x', and writes the CSV files distributed with this record. Because the dataset is generated from the published pages rather than retyped, it cannot silently drift from them.

The sectional callout table is the one exception: those item numbers were read off the manufacturer's own sectional drawings by a human, sheet by sheet. They are reported per sheet rather than merged, because the sheets are not supersets of one another - item 319 appears on the STX and MTX sheets but not on the XLT-X sheet, and 360A only on XLT-X. Note also that the drawing labels that frame XLT-X while the manufacturer's prose calls it XLTX; the sheet labels here follow the drawing.

Fields

series

G196 or D Mark III - the manufacturer's own series name.

pattern

The published dimensional pattern the series is built to.

item_no

The item number (G196) or POS number (D Mark III) used for that part on a sectional drawing.

part

What the item number identifies.

ysm_ref_suffix

The manufacturer's own reference suffix for that part.

materials

Material options offered for that part.

confirmed_before_order

What has to be confirmed from the nameplate or drawing before that part is ordered.

family

The frame (STX, MTX, LTX, XLTX) or group (1, 2, 3) a set of sizes belongs to.

sizes

The size designations that family covers, as discharge x suction - nominal impeller diameter.

sheet

Which sectional sheet a callout appears on.

pump_size

A size designation.

casing_ysm_ref / impeller_ysm_ref

The manufacturer's references for the casing and impeller of that size.

Table 1 - Item Numbers

19 rows. File: ysm-b73-item-numbers.csv

Series	Item no.	Part	YSM ref.	Materials offered
G196	100	Casing	100	WCB carbon steel, CF8M (316SS), CD4MCu duplex, CN7M (Alloy 20), Monel, Nickel, Hastelloy B and C, Titanium, Zirconium alloy
G196	101	Impeller (open)	3	CF8M, CD4MCu, CN7M (Alloy 20), Monel, Nickel, Hastelloy B and C, Titanium, Zirconium alloy
G196	184	Stuffing box cover (packed box)	106	WCB, CF8M, CD4MCu, CN7M, Monel, Nickel, Hastelloy, Titanium
G196	184M	Seal chamber (mechanical seal)	106B, 106F	WCB, CF8M, CD4MCu, CN7M, Monel, Nickel, Hastelloy, Titanium
G196	108	Frame adapter	108	Ductile iron
G196	228	Bearing frame	119	Cast iron; ductile iron for STX
G196	134	Bearing housing	201	Per YSM drawing, stated at quotation
G196	241	Frame foot	109	Per YSM drawing, stated at quotation
G196	122	Shaft	105, 105S	SAE 4140, 316SS, CN7M, Monel, Nickel, Hastelloy, Titanium
G196	126	Shaft sleeve	177	316SS, CN7M, Monel, Nickel, Hastelloy, Titanium
D Mark III	100	Casing	100	WCB carbon steel, CF8M (316SS), CF3M (316L), CD4MCu duplex, super duplex, CN7M (Alloy 20), Monel, Nickel, Inconel, Hastelloy B and C, Titanium, Zirconium alloy
D Mark III	103	Reverse-vane impeller	3	CF8M, CF3M, CD4MCu, super duplex, CN7M, Monel, Nickel, Inconel, Hastelloy B and C, Titanium, Zirconium alloy
D Mark III	106	Rear cover plate		WCB, CF8M, CF3M, CD4MCu, super duplex, CN7M, Monel, Nickel, Inconel, Hastelloy, Titanium
D Mark III	108	Bearing housing adapter	108	Ductile iron
D Mark III	119	Bearing housing	119	Ductile iron
D Mark III	201	Bearing carrier	201	Per YSM drawing, stated at quotation
D Mark III	109	Bearing housing foot	109	Per YSM drawing, stated at quotation
D Mark III	105	Shaft	105, 105S	SAE 4140, 316SS
D Mark III	177	Hook sleeve	177	Wet-end alloy of the pump

Table 2 - Frame and Group Families

6 rows. File: ysm-b73-families.csv

Series	Family	Sizes covered
G196	STX	1x1.5-6, 1.5x3-6, 2x3-6, 1x1.5-8, 1.5x3-8 and the 1x1.5-8LF low-flow size
G196	MTX / LTX	8 in, 10 in and 13 in sizes from 2x3-8 to 4x6-13, including 3x4-8G, 3x4-10H, 4x6-10H and the LF low-flow sizes. MTX and LTX share casing, impeller and cover; the power-end parts differ.
G196	XLTX	6x8-13, 8x10-13, 6x8-15, 8x10-15, 8x10-15G
D Mark III	Group 1	1.5x1-6, 3x1.5-6, 3x2-6, 1.5x1-8, 3x1.5-8 and the 1.5x1LF-4 and 1.5x1LF-8 low-flow sizes
D Mark III	Group 2	8 in, 10 in and 13 in sizes from 3x2-8 to 6x4-13A, including the H and HH high-head hydraulics and the 2x1LF-10 and 3x1.5LF-13 low-flow sizes
D Mark III	Group 3	8x6-14A, 10x8-14, 6x4-16, 8x6-16, 8x6-16A, 10x8-16, 10x8-16H, 10x8-17

Table 3 - Sectional Sheet Callouts

104 rows. File: ysm-b73-sectional-callouts.csv

Series	Sheet	Item numbers called out
G196	STX sheet (sheet 1)	100, 101, 112, 122, 126, 136, 168, 184M, 228, 250, 319, 332A, 333A, 351, 358A, 370, 418, 496, 496A
G196	MTX sheet (sheet 2)	100, 101, 108, 112, 122, 126, 136, 168, 184M, 228, 250, 319, 332A, 333A, 351, 358A, 360, 370, 418, 469B, 496, 496A
G196	XLTX sheet (sheet 3, XLTX frame)	100, 101, 108, 112, 122, 126, 136, 168, 184M, 228, 250, 332A, 333A, 351, 358A, 360, 360A, 370, 418, 496, 496A
D Mark III	Exploded view (sheet 1)	100, 103, 104, 105, 106, 107, 108, 109, 109A, 110, 111, 111A, 112, 113, 114, 115, 115A, 118, 119, 120, 121, 124, 125, 129, 130, 131, 133, 134, 135, 136, 139, 140, 153, 190, 190G, 200, 201, 201A, 201B, 201C, 201D, 201E

Table 4 - Size Reference

60 rows. File: ysm-b73-size-reference.csv

Series	Pump size	Casing YSM ref.	Impeller YSM ref.
G196	1x1.5-6	GAB100	GAB3
G196	1.5x3-6	GAD100	GAD3
G196	2x3-6	GKE100	GKE3
G196	1x1.5-8	GAC100	GAC3
G196	1x1.5-8LF	GYB100	GYB3
G196	1.5x3-8	GAE100	GAE3
G196	2x3-8	GAF100	GAF3
G196	3x4-8	GAG100	GAG3
G196	3x4-8G	GAGG100	GAGG3
G196	1x2-10	GAH100	GAH3
G196	1x2-10LF	GYC100	GYC3
G196	1.5x3-10	GAJ100	GAJ3
G196	2x3-10	GAL100	GAL3
G196	3x4-10	GAN100	GAN3
G196	3x4-10H	GAP100	GAP3
G196	4x6-10	GAQ100	GAQ3
G196	4x6-10H	GAR100	GAR3
G196	1.5x3-13	GAS100	GAS3
G196	1.5x3-13LF	GYD100	GYD3
G196	2x3-13	GAT100	GAT3
G196	3x4-13	GAU100	GAU3
G196	4x6-13	GAW100	GAW3
G196	6x8-13	GNA100	GNA3
G196	8x10-13	GNB100	GNB3
G196	6x8-15	GNF100	GNF3
G196	8x10-15	GNC100	GNC3
G196	8x10-15G	GND100	GND3
D Mark III	1K1.5x1LF-4	DYA100	DYA3
D Mark III	1K1.5x1-6	DKA100	DKA3
D Mark III	1K3x1.5-6	DKC100	DKC3
D Mark III	1K3x2-6	DKE100	DKE3
D Mark III	1K1.5x1LF-8	DYB100	DYB3
D Mark III	1K1.5x1-8	DKB100	DKB3
D Mark III	1K3x1.5-8	DKD100	DKD3
D Mark III	2K3x2-8	DAF100	DAF3
D Mark III	2K4x3-8	DAG100	DAG3
D Mark III	2K2x1LF-10	DYC100	DYC3
D Mark III	2K2x1-10A	DAH100	DAH3
D Mark III	3x1.5-10	DAJ100	DAJ3
D Mark III	2K3x1.5-10A	DAK100	DAK3
D Mark III	3x2-10	DAL100	DAL3
D Mark III	2K3x2-10A	DAM100	DAM3
D Mark III	2K4x3-10	DAN100	DAN3
D Mark III	2K4x3-10H	DAP100	DAP3
D Mark III	2K6x4-10	DAQ100	DAQ3

Table 4 - Size Reference (continued)

Series	Pump size	Casing YSM ref.	Impeller YSM ref.
D Mark III	2K6x4-10H	DAR100	DAR3
D Mark III	2K3x1.5LF-13	DYD100	DYD3
D Mark III	2K3x1.5-13	DAS100	DAS3
D Mark III	2K3x2-13	DAT100	DAT3
D Mark III	2K4x3-13	DAU100	DAU3
D Mark III	2K4x3-13HH	DAV100	DAV3
D Mark III	2K6x4-13A	DAW100	DAW3
D Mark III	3K8x6-14A	DNA100	DNA3
D Mark III	3K10x8-14	DNB100	DNB3
D Mark III	3K6x4-16	DNE100	DNE3
D Mark III	8x6-16	DNF100	DNF3
D Mark III	3K8x6-16A	DNG100	DNG3
D Mark III	3K10x8-16	DNC100	DNC3
D Mark III	3K10x8-16H	DND100	DND3
D Mark III	3K10x8-17	DNH100	DNH3

About This Record

Publisher

Jinan Yingsiman Machinery Co., Ltd., trading as YSM Pumps, manufactures ANSI/ASME B73.1 chemical process pumps and non-OEM replacement pumps and wet-end parts built to the same dimensional patterns, plus ISO 5199/2858, API 610 OH2, multistage, vertical turbine and paper stock pumps. The company was registered in Jinan in 2018; its works in Zibo, Shandong has been in operation since 2010.

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Sources these tables were compiled from

Goulds 3196 pattern parts, with materials: <https://ysmpumps.com/goulds-3196-pump-parts/>

Durco Mark 3 pattern parts, with materials: <https://ysmpumps.com/durco-mark-3-pump-parts/>

Replacement pumps and parts programme: <https://ysmpumps.com/replacement-ansi-pump-parts/>

Summit 2196 replacement, size coverage: <https://ysmpumps.com/product/summit-2196-replacement/>

How a B73.1 size designation is built: <https://ysmpumps.com/resources/ansi-pump-size-codes/>

Corrections

Errors can be reported to the publisher; corrections are issued as a new version of this record rather than by editing the published files. Any newer version supersedes this one.

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